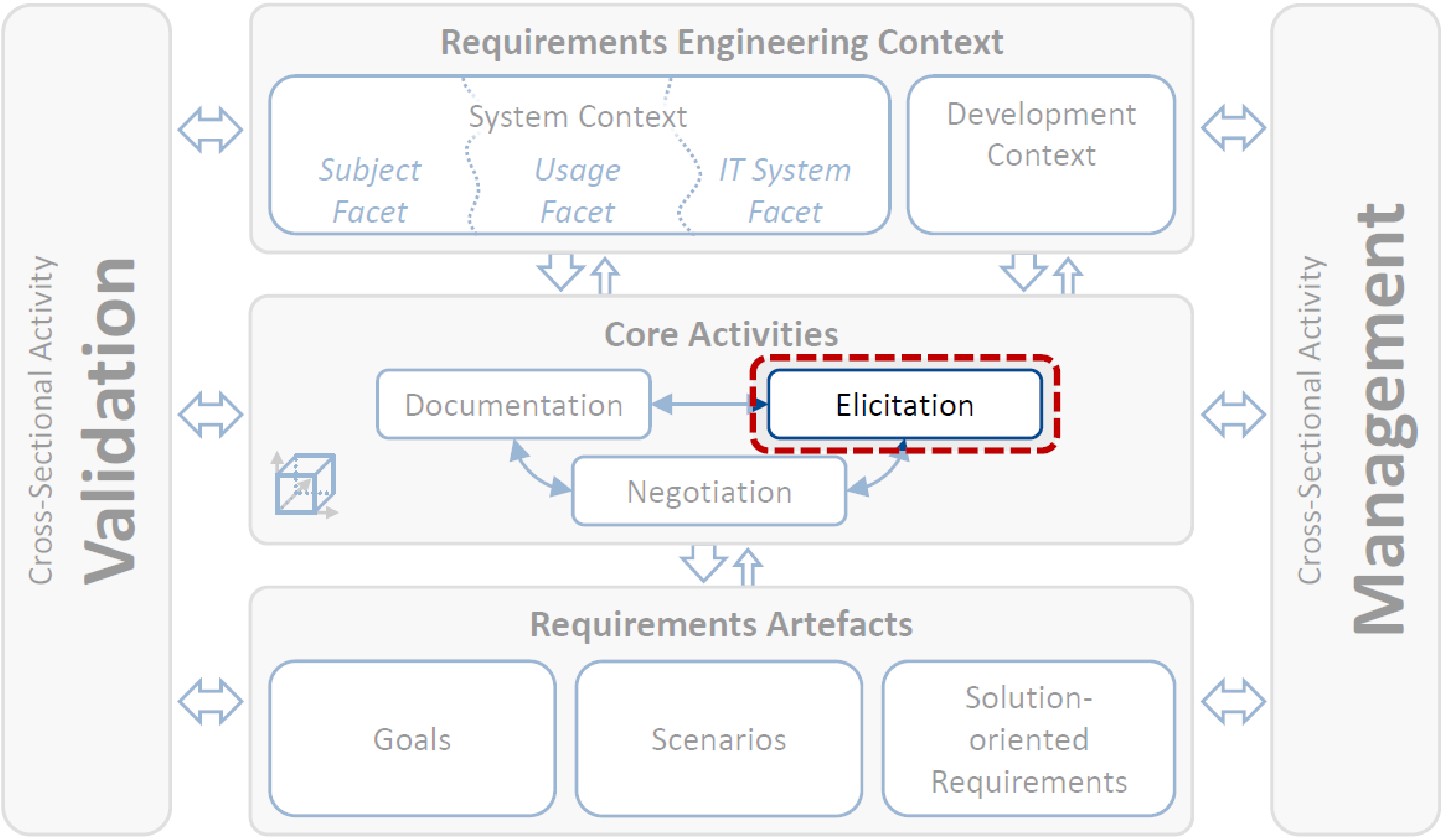
CKS121/CSE 242 SOFTWARE REQUIREMENTS

Elicitation (Part 1)

UNIVERSITI SAINS MALAYSIA
SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

Requirements Engineering Framework



LEARNING OUTCOMES

The students

- understand the goal of requirements elicitation
- learn the different requirement sources.
- learn how to identify requirement sources and assess their relevance.
- understand the difference between elicitation of existing requirements and creation of innovative requirements.
- get an overview on common elicitation techniques.

Structure of this Lecture

- **Introduction to Requirements Elicitation**
- Requirements Sources
- Identification of Requirements Sources
- Eliciting Existing Requirements vs. Creating Innovative Requirements
- Common Elicitation Techniques

Goal of Requirements Elicitation

D Requirements elicitation is a core requirements engineering activity. The goal of requirements elicitation is threefold:

(1) **Identify** relevant **requirements sources**.

(2) Elicit **existing requirements** from the identified sources.

(3) Develop **new and innovative requirements**.

Revisit!

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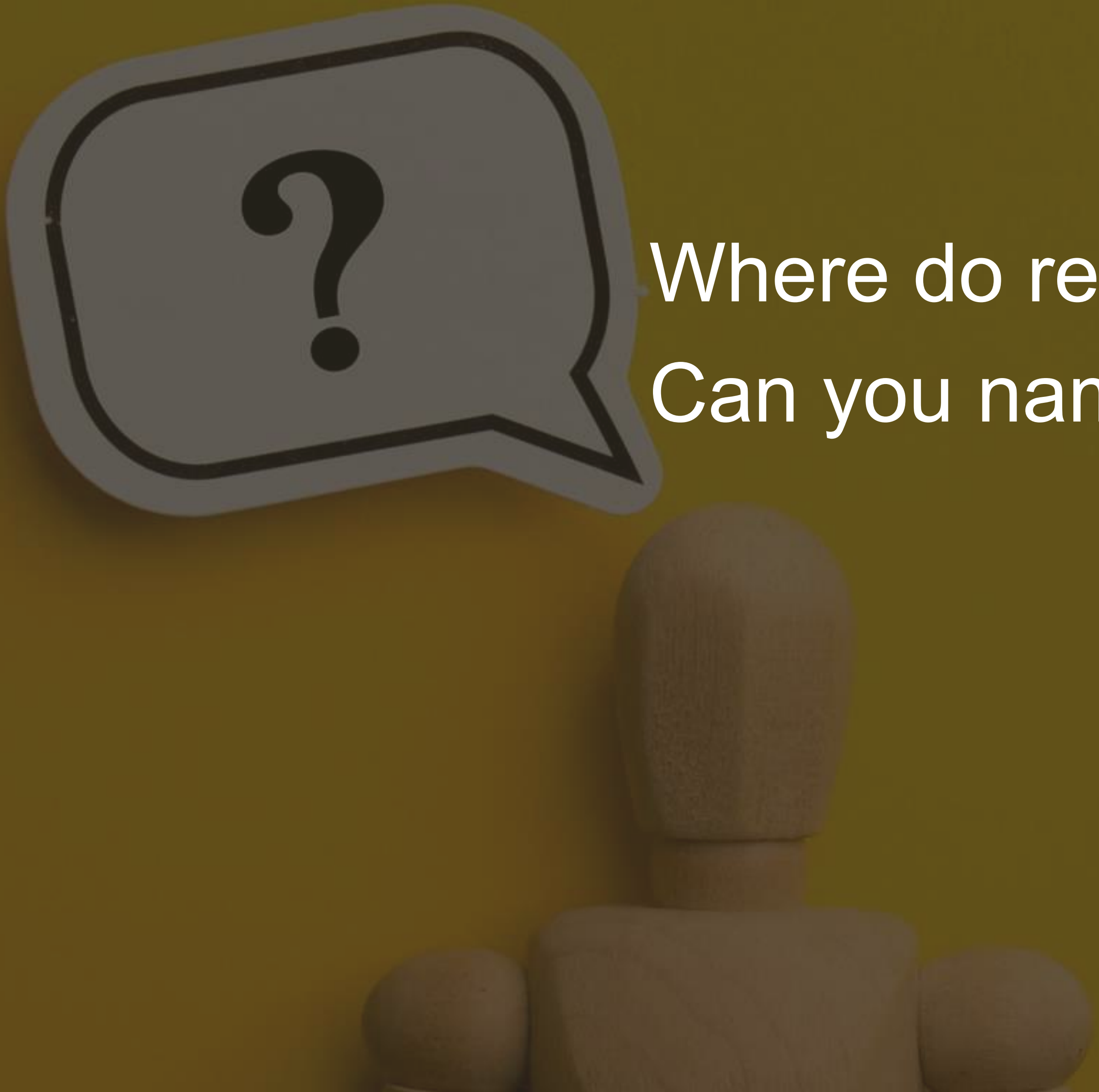
Importance

- Some requirements sources might be obvious or might have already been identified.
  Nevertheless many requirement sources are typically unknown (at the beginning of requirements engineering).
- Not identifying or not considering relevant requirement sources during requirements engineering typically leads to
  - incomplete requirements (e.g. requirements are overlooked).
  - insufficient agreement or unrecognized conflicts about requirements.
  - At the latest, overlooked requirements or insufficient agreement lead to change requests during system operation (and higher costs)!
- Identifying all relevant requirements sources is thus essential!



Where do requirements come from?

Can you name possible sources?

Three Types of Requirements Sources

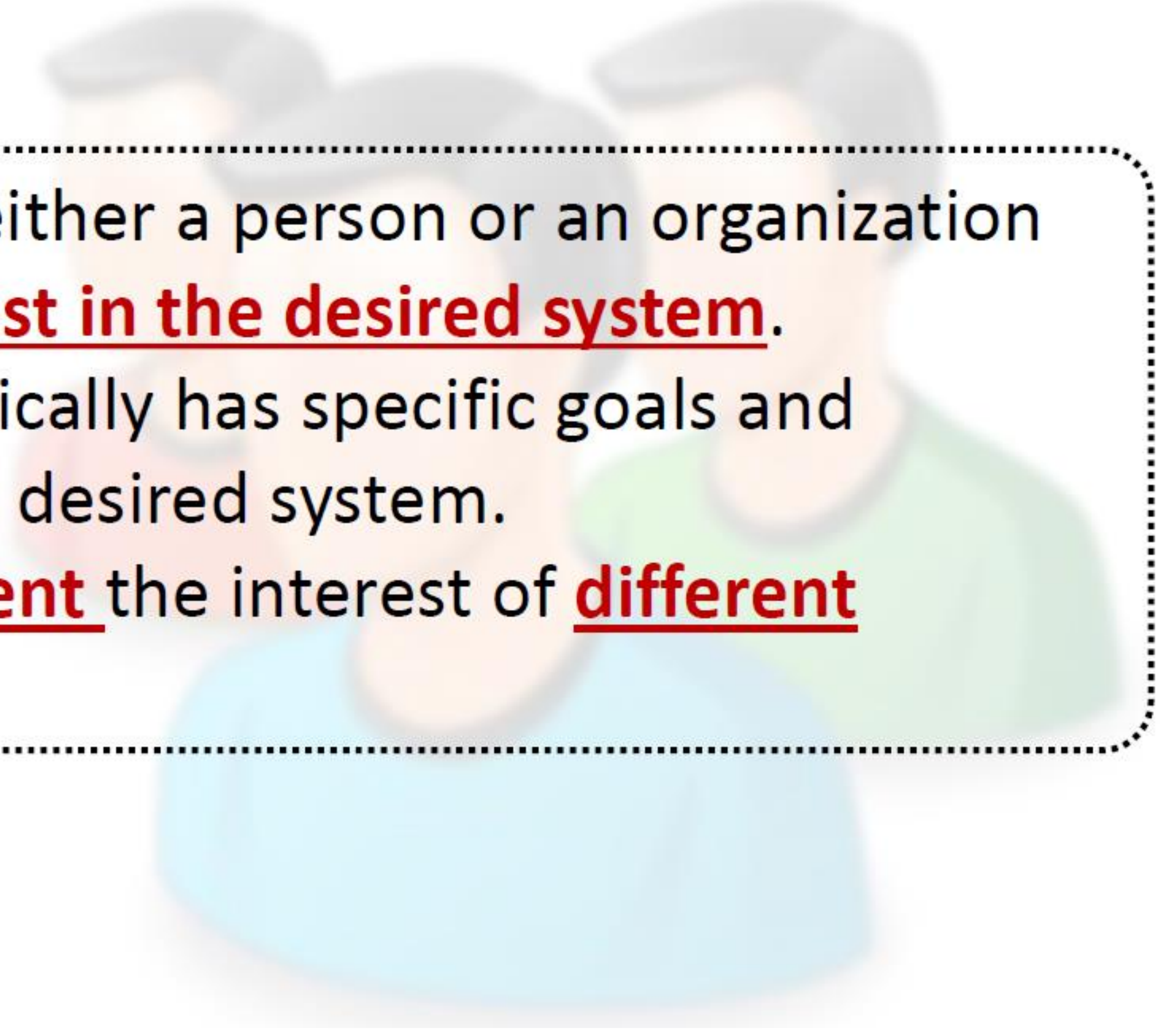
Existing Systems

Documents

Stakeholders

All icons from [1]

Stakeholder (1)



D A **stakeholder** is either a person or an organization with **potential interest in the desired system**.
Each stakeholder typically has specific goals and requirements for the desired system.
A person can **represent** the interest of **different stakeholders**.

Stakeholder (2)

- A **stakeholder is a context object**
- A **stakeholder can belong to**
  - any facet of the system context (subject, usage or IT system),
  - the development context,
  - the requirements engineering context.

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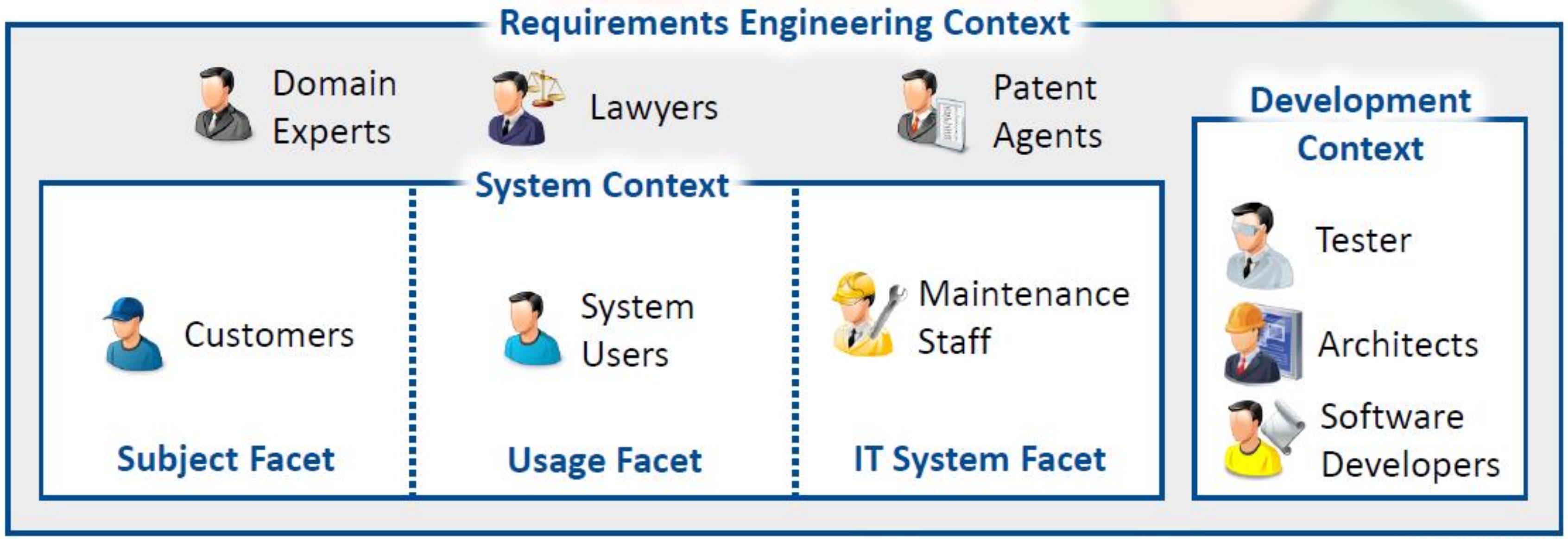
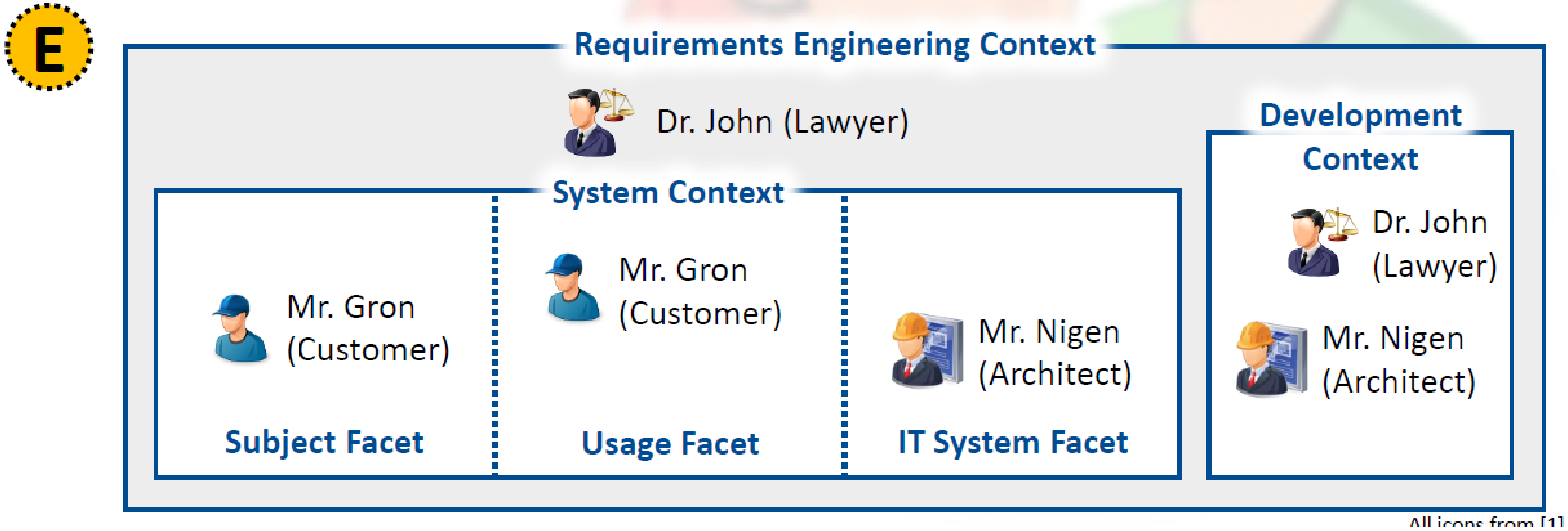


All icons from [1]

Stakeholder (3)

- A stakeholder typically has **knowledge about several context objects** and their relationships.
- A Stakeholders can have **knowledge about different facets of the system context** as well as the **development context**.
  I.e. a stakeholder can be relevant for eliciting requirements about different context compartments.



Documents (1)

Existing documents contain relevant information to be considered when defining the requirements for the desired system.

We differentiate between three types of potentially relevant documents:

1. **General binding documents**
   This type of documents subsume documents defined by standardization bodies, governments, special interest groups, trade or professional organizations or the like.
   Examples for general binding documents include laws and standards.

Documents (2)

2. **Organization-specific documents.**
   This type of documents are defined within an organization involved in the development or use of the desired system.

   Examples for organization-specific documents include development guidelines, product strategies, HCI guidelines, IT-strategies, security guidelines, business process documentation or market analysis reports.

3. **Product-/System-specific documents** are documents describing any kind of development artefact (e.g., requirements, code, architecture, test cases, use cases) of a predecessor system or similar system.

   Examples for development documents include change requests documents, error reports, user manuals, system architecture documents, requirements specifications, maintenance documents, test documentation and marketing material.

Existing Systems (1)

Existing systems provide an excellent source for detecting and uncovering existing requirements:

- **Requirements realized by the existing system** might still be **relevant for the new system**.
- Existing systems should be analysed
  - **to elicit** the properties of the existing system.
  - **to identify required enhancements** of the existing system.
  - **to identify** known **deficiencies** of the existing system which should be avoided in the future.
  - **to identify previous errors** already fixed in the predecessor system and thus to avoid making the same errors in the desired system.
- **Stakeholders using or involved** in the development or operation of existing systems are additional **potential valuable requirements sources**.

Existing Systems (2)

We differentiate between three types of systems:

1. **Predecessor systems:**
    - Predecessor (legacy) systems typically have a similar purpose as the system to be developed. They are still in use within an organization (or have been in use) and are typically (partial) replaced by the new system
2. **Systems of competitors:**
    - Analysing systems competing with the system to be developed is essential to not oversee important requirements and to ensure the system offers some differentiating features for its users.
3. **Systems from other domains:**
    - Systems developed for another domain might offer properties, use innovative technology or offer unique features relevant for the desired system. Such systems could be valuable sources for innovative requirements for the desired system.

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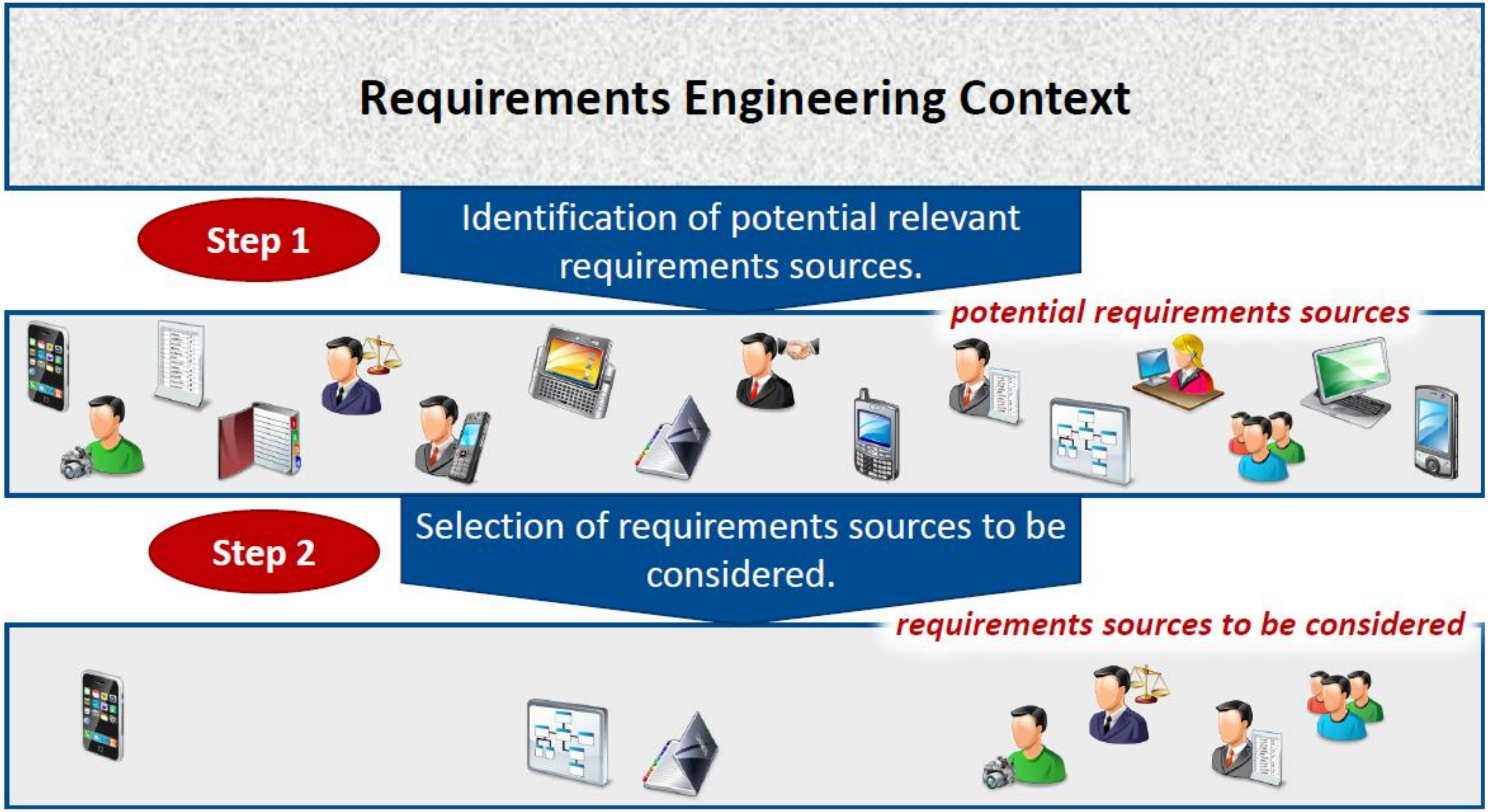
- **Identification of Requirements Sources**
- Eliciting Existing Requirements vs. Creating Innovative Requirements
- Common Elicitation Techniques

Identification of Requirements Sources

Two-Step Procedure



Step 1 –Potential Relevant Requirements Sources

Goal: Identify a large set of potential relevant requirements sources.

Activities:

1.1 Use already known requirements sources and/or suitable checklists to **identify additional**, potential requirement **sources**.

1.2 **Record newly identified**, potential relevant requirement sources.

1.3 For each newly identified requirement source, **perform activity 1.1 again**.

Iterate until the set of identified sources becomes (more or less) stable.

Step 1

Step 1: Hints

- Consider all **parts of the requirements engineering context** for identifying potential requirement sources.
- First, **identify relevant context objects**. Then identify requirements sources potentially having **knowledge about those objects**.
- Consider all **three types of requirement sources** (stakeholders, documents and existing systems).
- Use **domain and system-specific checklists** to support the identification of requirements context objects and requirement sources.

Template for checklist	Usage Facet	Subject Facet	IT System Facet	Development Context	RE Context
Stakeholders	...	...	...	...	...
Documents	...	...		...	...
Systems	...	...	...	...	...

Step 1: Checklist for Context Object

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Checklist for a car safety system

Usage Facet
- driver
- ACC-function
- maintenance staff
- engine control
- acceleration system

Subject Facet
- pedestrians
- driver
- vehicles in front
- streets
- objects on the streets

IT System Facet
- OSEK (operating system)
- Acceleration sensor
- Rotation sensor
- FlexRay communication system
- car components

ACC: Adaptive Cruise Control
OSEK: **O**ffene **S**ysteme und deren Schnittstellen für die **E**lektronik im **K**raftfahrzeug
HCI: Human Computer Interaction

Development Context
- car engineer
- sensor expert
- HCI expert
- software architect

Requirements Engineering Context
- regulatory agencies
- flight safety systems
- car producer
- legal expert

Step 1: Checklist for Requirements Sources

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Checklist for a car safety system

Stakeholders	Documents	Stakeholders
• car driver • professional driver • accident assessor • physician • car technician • car engineer • maintenance staff • sensor expert • engineer • control unit display • safety experts • regulatory agencies • lawyers	• car manual • user interface descriptions • documents of the engine control • specification of analogue systems • AUTOSAR standard	• previous safety system • safety systems of competitors • sensors systems • HCI systems • systems interacting with safety system • flight safety systems • train safety systems • metro safety systems

Step 2 – Selection Requirements Sources (1)

Goal: Select the most relevant requirements sources from the identified sources.

Execute the so called **100 Dollar Test** in a group meeting:

- In a 100-dollar test **each stakeholders metaphorically spends 100 dollars** on the items (requirements sources) to be assessed.
- Each stakeholder individually **distributes the money** to the **requirements sources** she/he thinks are **relevant.**
- The **amount of money** a specific requirements source receives **defines the relative weighting** of this source.
- The group **selects** the **highest weighted requirements sources to be considered first**.
  Remaining sources might be considered later.

Step 2

Step 2 - Hints for Selecting Most Relevant Sources

- The cut-off point for considering identified requirement sources depends on the project setting, e.g. on the time available, the budget available and the availability of the requirements sources.
- Check if selected sources cover all relevant parts of the RE context.
- You might use the 100-dollar test to select the requirements source separately for each part of the requirements engineering context.
- When selecting the stakeholders for the assessment, ensure an adequate representation of the RE context.

Step 2

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Two Important Kinds of Requirements

Besides identifying requirements sources, the goal of requirements elicitation is twofold:

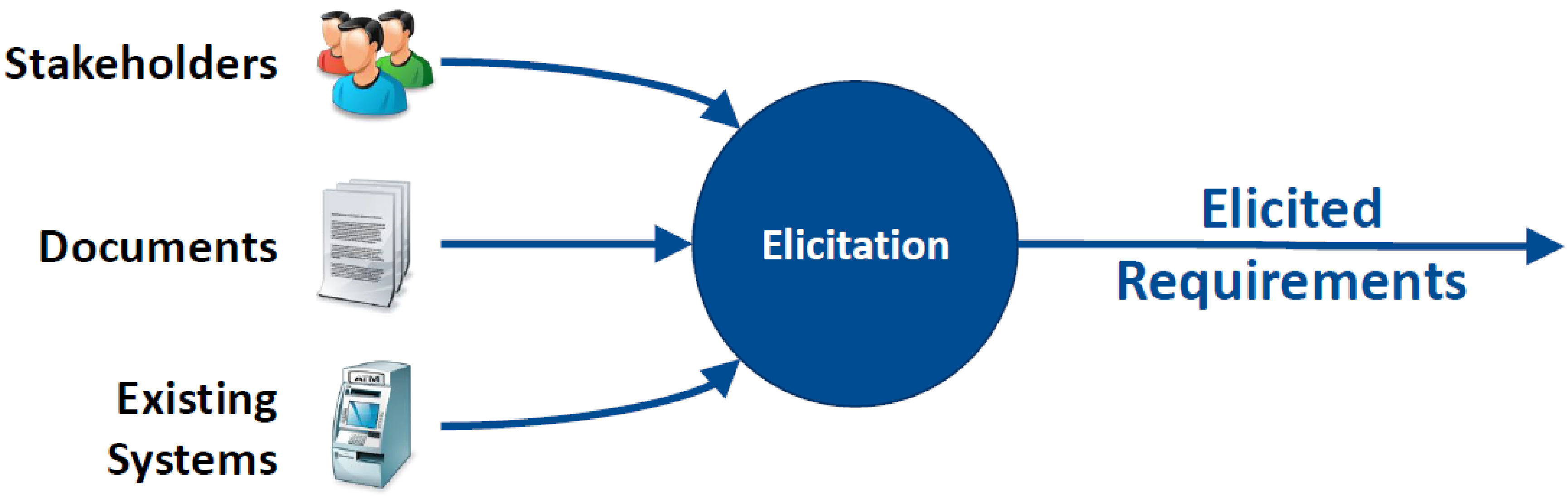
(1) **Elicit existing requirements** from relevant requirements sources

(2) **Create** and develop **innovative** (new) **requirements**.

Existing and innovative requirements are equally important for system success!

Eliciting Existing Requirements

Stakeholders can elicit requirements from the three types of requirements sources:



Elicitation Existing Requirements from Stakeholders

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Stakeholders express their requirements guided by a **questionnaire**:

Question 12: How can the safety of a car during winter be improved?

Answer: The car should displays a warning when the outside temperature is below 3°C to indicate a high probability of icy roads.

Question 13: In your opinion, how can the risk of rear-end collisions be decreased?

Answer: The safety system should warn the driver if the distance to the vehicle in front gets critically low – and even might initiate an emergency breaking if required.

Elicitation Existing Requirements from Documents (1)

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> §
> Regulation 2010/156/EC
> […] All electronic systems in a vehicle that directly or indirectly influence the occupants' safety or the safety of other traffic participants must be designed in such a way that failure of the electronic system has no negative effects on safety. […]

The **law** (regulation 2010/156/EC) **must be considered** for the safety system of the car.

Based on the law the following goals might be defined:

- G1: The driver shall be able to override the system actions at any time.
- G2: The system must not disturb any other system even in the case of a system failure.

Elicitation Existing Requirements from Documents (2)

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Error FA-2003-1-10-F3:	The motor heat emission heats the sensor responsible for measuring the outside temperature. Therefore, the displayed outside temperature is incorrect (too high), especially when driving at low speed.
Error Correction:	The sensor was put in a new position which protects the temperature sensor from the engines heat emission (see Correction Report K-B-2003-4-12-k5).

Based on the error report, the stakeholder might derive the following requirement:

Req-15: Protect the temperature sensor from the heat emission of the engine.

Elicitation Existing Requirements from Existing Systems (1)

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To elicited requirements from existing systems, typically the following two steps are executed:

(1) The **incarnation of the existing system** is elicited and defined resulting in an **incarnation model** of the predecessor system.

(2) Based on the modelled incarnation of the system an **essential model** is created by **abstracting from the incarnation**.

Elicitation Existing Requirements from Existing Systems (2)

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Excerpt

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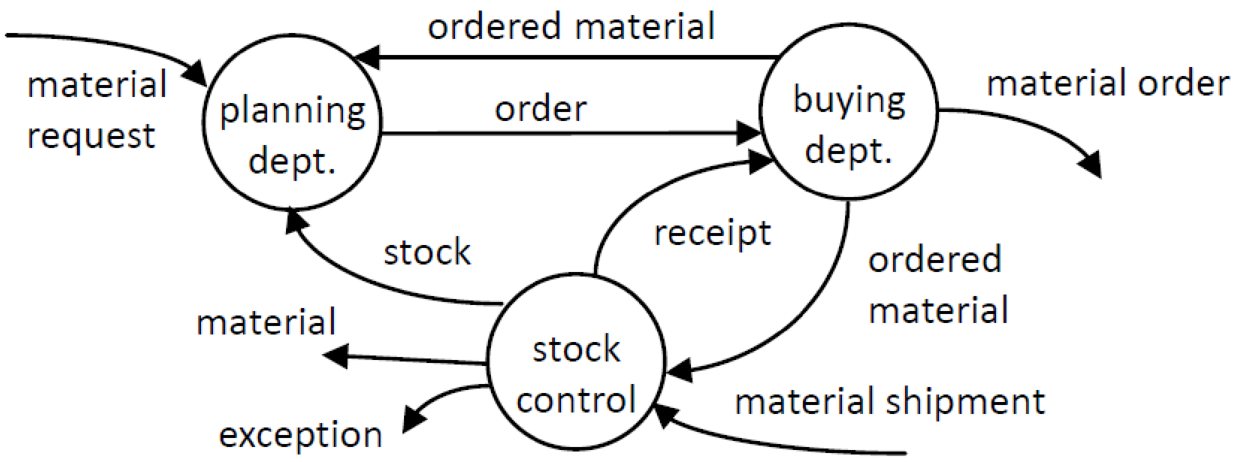
Incarnation of the predecessor ERP system



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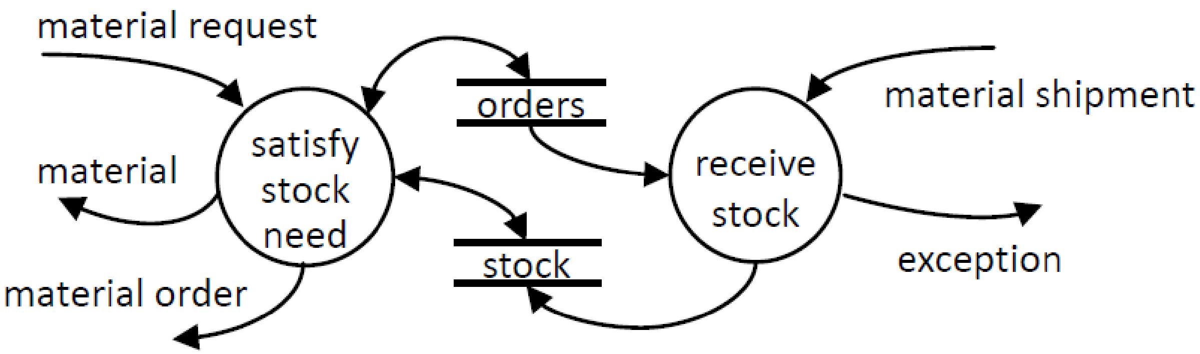
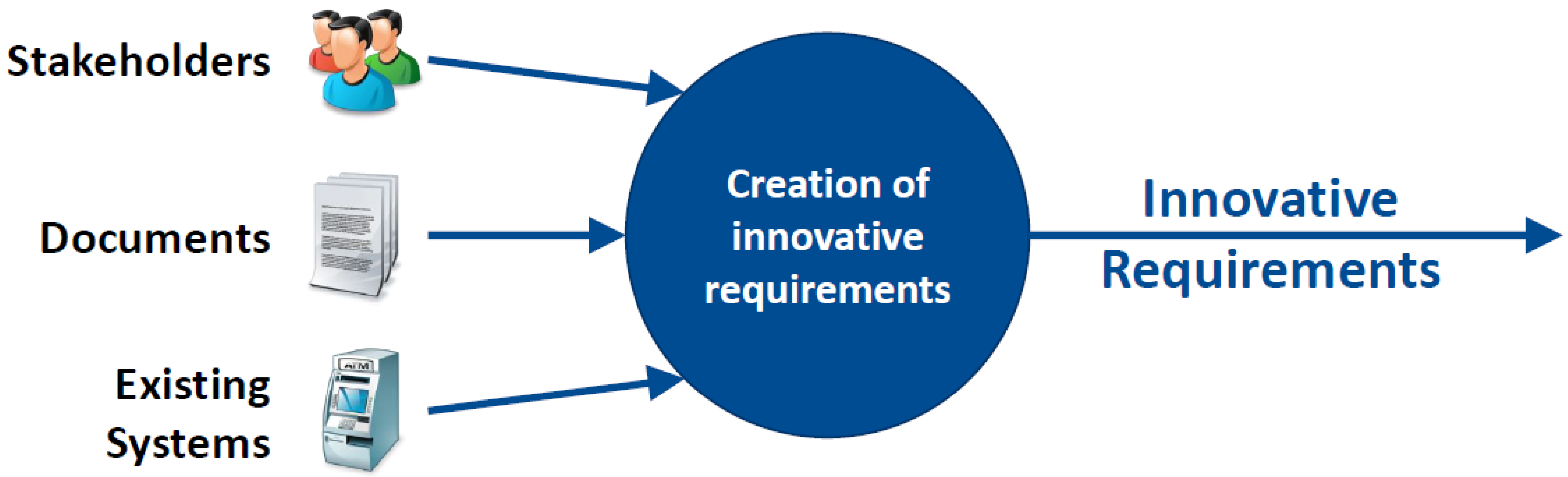
Essence of the predecessor ERP system



Creation of Innovative Requirements (1)

- Innovative (new) requirements **cannot** be elicited from requirements sources in the same way as existing requirements!
- Innovative requirements have to be created in **creative processes** using creativity techniques such as brainstorming.



Creation of Innovative Requirements (2)

Creativity is

> "the **ability to produce work** that is both **novel** (i.e. original unexpected) and **appropriate** (i.e. useful, adapted to a task constraints)" [Sternber and Lubart 1999]

Three Categories of Creativity:

- **Exploratory** creativity
  - Search space of partial and complete possibilities.
- **Combinational** creativity
  - Make unfamiliar connections between familiar possibilities in the search space.
- **Transformational** creativity
  - Challenge the constraints on the search space.
  - Enlarge space of possible ideas to explore.

Creation of Innovative Requirements with Stakeholders

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- During brainstorming for “What to display in the next generation of head up display for a car”, the following ideas have been collected:
  - Traffic signs
  - Circumnavigate traffic jam
  - Navigation instructions
  - Proactive traffic light monitoring
  - Current speed
  - Slippery road warning
- The idea to monitor the traffic light status based on a proactive traffic light monitoring is innovative and considered in the next system release.

Creation of Innovative Requirements using Existing Documentation

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The overall vision is to build a **smart refrigerator** which knows the products stored and offers innovative functions based on this knowledge.

- To identify some possible innovative requirements for the fridge, the requirements engineer analyses the specification of a food trader's ERP system.
- He identifies the functionality to monitor the "best before" dates as potential innovative requirements for the fridge.
- Similar as the ERP system, the fridge should control the "best before" dates and provide the information to the user – to ensure food is used before it spoils.

Creation of Innovative Requirements using Existing Systems

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To identify some potentially innovative requirements for a **smart refrigerator** the stakeholder decide to participate in a **demonstration of smart televisions**.

During the demonstration and by talking to the television experts the stakeholder **identify features** realized in the smart television system potentially useful and **innovative for the smart refrigerator**. The refrigerator shall interact with the smart phones of the consumers in order to **suggest recipes** corresponding to their diet and the available food. Furthermore, the refrigerator shall **create a shopping list** according to the available products and send them fully autonomous to the smart phones.

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Elicitation Techniques and Assistance Techniques

- Elicitation techniques support
  - the elicitation of existing requirements.
  - the creation of innovative requirements.
  - to elicit requirements sources.
- Assistance techniques support the different elicitation techniques, e.g., by developing new ideas or experiencing feature of the future system.
- We first introduce common elicitation techniques; than common assistance techniques.

Six Common Requirements Elicitation Techniques

- **Interview**
  Elicitation of requirements and context information from a stakeholder or a group of stakeholders.
- **Workshop**
  A group of stakeholders develops requirements for a system.
- **Focus Groups**
  A group of stakeholders focus on a specific item to identify the requirements regarding this item.
- **Observation**
  An observer elicits requirements by observing stakeholders or existing systems.
- **Questionnaire**
  A stakeholder writes down his requirements by answering predefined questions.
- **Perspective-based Reading**
  A stakeholder reads a document from a previously defined perspective, e.g., from the perspective of a user or from the perspective of a tester.

Common Elicitation Techniques

Rough Classification

Technique	Effort	Suited for...		
		Identifying requirement sources	Eliciting existing requirements	Developing innovative requirements
Interview	Medium to high	X	X	(X)
Workshop	High to very high	X	X	X
Focus groups	Medium to high		X	X
Observation	High to very high		X	
Questionnaire	Low to medium	X	X	
Perspective-based reading	Medium to high		X	

Template for Descripting Elicitation Techniques

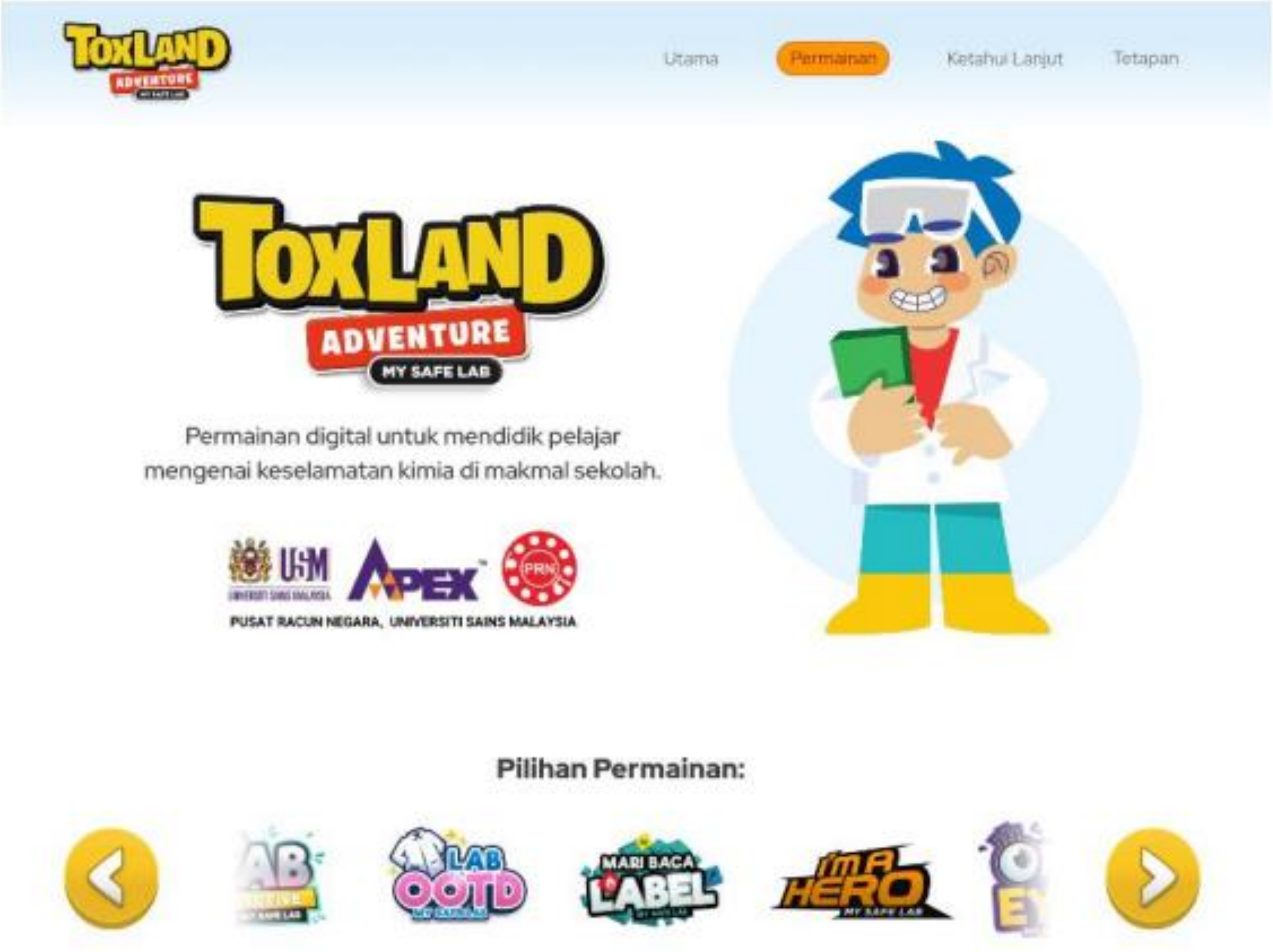
Application

1. **Preparation** (common/specific)
   Necessary actions for preparing the execution of the technique.
2. **Execution**
   Essentials of performing the technique.
3. **Follow-Up**
   Necessary actions to be executed after having performed the technique.
4. **Critical success factors**

Key characteristics

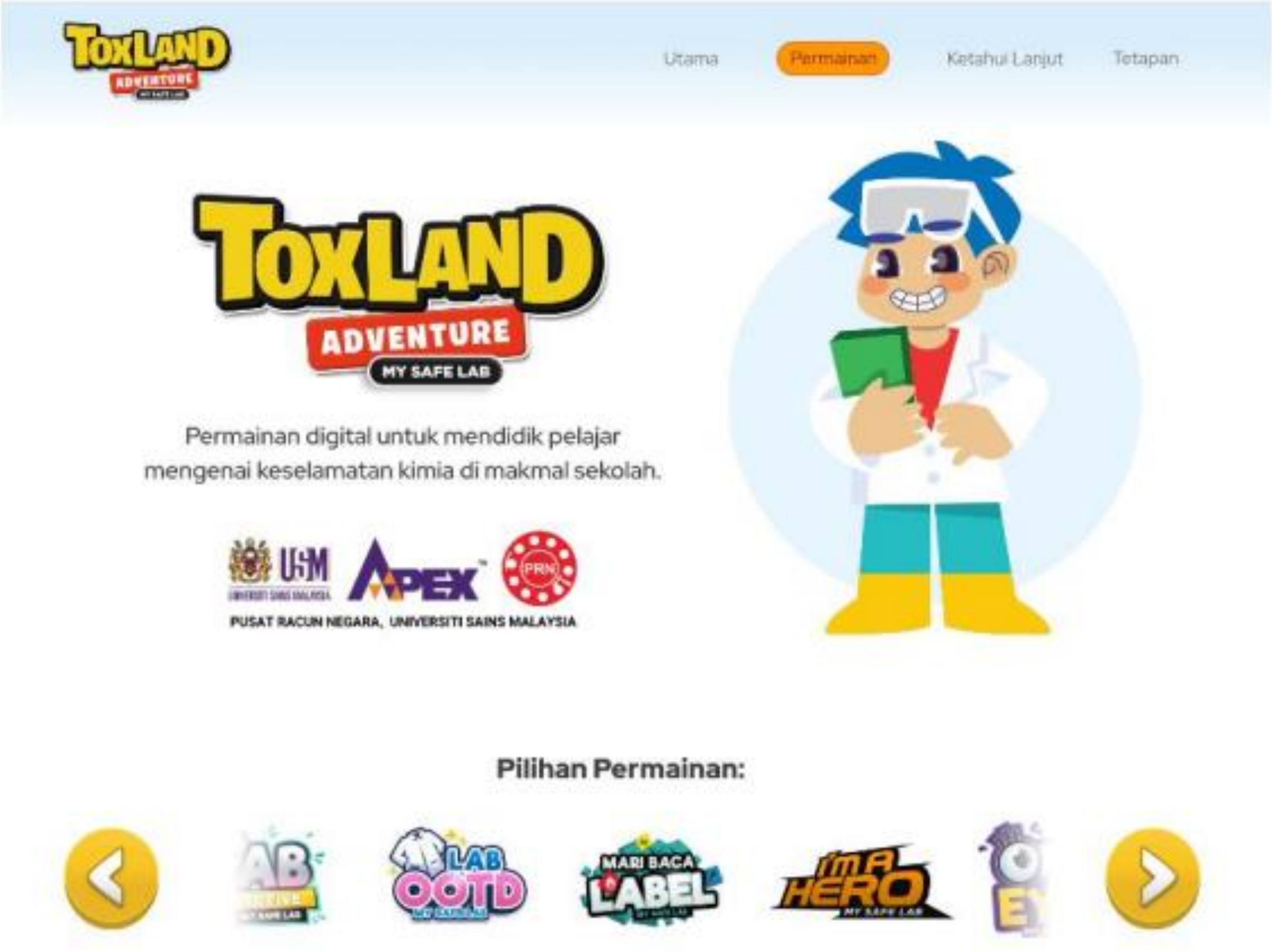
5. **Benefits** for the requirements engineering activity.
6. **Effort** as rough estimation for the application of the technique.

CASE STUDY 1: TOXLAND



- Pusat Racun Negara (PRN) [National Poison Centre] is promoting awareness and understanding of the dangers of chemical exposure among children and young adults.
- System objectives:
  - To make learning about toxicology more engaging via game learning
  - To design educational and entertaining games that help student to learn and retain important information about toxicology and chemical safety in a laboratory setting

CASE STUDY 1: TOXLAND



- Identify the requirements sources
- Identify the stakeholders
- Identify possible requirement elicitation techniques

that could help you to document the requirement specifications.

Summary

- The goal of requirements elicitation is to identify relevant requirements sources, elicit existing requirements from the identified sources and develop innovative (new) requirements.
- There are three types of requirement sources: documents, existing systems, and stakeholders.
- Two-step process of identifying requirement sources; (1) identification of potential relevant requirements sources; (2) selection of the sources to be considered.
- Missing requirement sources leads to incomplete specifications and low quality systems.
- Common elicitation techniques are used to elicit existing requirements and to develop innovative requirements. They are supported by specific assistance techniques.

Questions for Self-Assessment

- What are the goals of requirements elicitation?
- Name the three types of requirement sources. Why is each source important?
- How are requirement sources identified?
- How can the relevance of requirements sources be assessed?
- What differences exist between elicitation of existing requirements and development of innovative requirements?